Zhantu Zhao

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EDUCATION

Xi'an Jiaotong-Liverpool University, China 09/2023 – 07/2027
BSc Applied Mathematics | **GPA**: 3.93/4.0 | Ranking: Top5% | GRE: 326

Core Courses: Multivariable Calculus | Advanced Linear Algebra | Introduction to Probability and Statistics | Analysis 1 & 2 | Introduction to Abstract Algebra | Dynamic Modeling | Introduction to the Methods of Applied Mathematics | Statistical Distribution Theory | Vector Fields: Theory and Applications | Ordinary Differential Equations and Control | Complex Functions | Introduction to Programming in Java

PUBLICATIONS AND PATENTS

- **Zhantu Zhao**, Xiangqian Sun. *Efficient Choice-Based Facility Location: A Hybrid AHP/DEA-Greedy Approach*. The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026). pp.15–26. (Oral Presentation). DOI: 10.1007/978-981-92-1947-6_2
- **Zhantu Zhao**, Xiangqian Sun. *Beyond Single-Positive: A Perturbed Fenchel-Young Loss for Top-K Recommendation*. ACM International Conference on Web Search and Data Mining (WSDM), 2027. Under Review
- **Zhantu Zhao**, Xiangqian Sun. *Multi-Positive Top-K Recommendation: A Differentiable Dynamic Programming Approach*. International Conference on Learning Representations (ICLR), 2027. Under Review
- **Zhantu Zhao**, Jin Lu, Ke Deng, Huaxin Zhao, Tong Wu. *InterSelf: Learning World Models under Hidden Action Execution*. International Conference on Learning Representations (ICLR), 2027. Under Review

AWARDS AND HONORS

- Academic Excellence Award (Top 5%) | XJTLU 07/2026
- Academic Excellence Award (Top 5%) | XJTLU 07/2025
- National Encouragement Scholarship | Ministry of Education of China 2025
- Academic Excellence Award (Top 5%) | XJTLU 07/2024
- SURF Poster Nomination Award (Top 15 / 400) | XJTLU 2024

RESEARCH EXPERIENCE

InterSelf-DreamerV3: World Models under Hidden Action Perturbations 04/2026 – Present
University of Georgia Summer Research Program | Advisor: Prof. Jin Lu

- Investigated whether a world model could recover hidden action effects when an agent's issued action differs from what the environment actually executes; extended the Transformer-based world model in DreamerV3 to settings with actuator noise, delay, or corruption; submitted a paper to **ICLR 2027**
  - Built five action-corruption modes (normal, sticky, lazy, left-right confusion, fire dropout) on Atari; turned a standard robustness test into an identifiability problem tied to unobserved action effects
  - Extended the RSSM with an explicit latent operator representing hidden action effects; embedded it directly into the imagination chain rather than treating it as a disconnected auxiliary predictor
  - Designed a three-tier ablation (E1 standard DreamerV3, E2 full InterSelf, E3 width- and parameter-matched baselines) to separate gains from hidden-action modeling versus added network capacity
  - Breakout showed consistent gains in standard comparison, but capacity-matched testing across Pong, Breakout, and SpaceInvaders revealed game- and seed-dependent variability
  - Managed large-scale RL experiments on university HPC cluster (RTX 4090 GPUs) using Slurm job scheduler, running dozens of parallel training jobs across multiple seeds and corruption modes; implemented SHA256-based version control for environment wrappers and model checkpoints to ensure reproducibility

QuotaRAG: Adaptive Retrieval under a Global Context-Token Budget 03/2026 – 08/2026
Summer Undergraduate Research Fellowship (SURF) | Advisor: Prof. Xiangqian Sun

- Reframed retrieval depth in Retrieval-Augmented Generation (RAG) as a workload-level resource allocation problem; determined how to distribute a limited context-token budget across queries of varying difficulty to maximize overall generation quality
  - Estimated generator-aware marginal utility for each retrieval prefix; linked the decision to add another passage directly to expected generation improvement rather than relying solely on retriever relevance scores
  - Applied a Prefix Lagrangian formulation with a deterministic budget-repair step; converted the global constraint into per-query decisions while keeping token consumption within budget
  - Observed an approximately 15% relative improvement on MuSiQue in early experiments, then re-verified this gain against a strong adaptive-K baseline before attributing it to the workload-level allocation mechanism

UR-RankTopK: Structured Multi-Positive Learning for Top-K Recommendation 03/2026 – 08/2026

Summer Undergraduate Research Fellowship (SURF) | Advisor: Prof. Xiangqian Sun

- Explored training recommendation models on a fixed-cardinality set of multiple positive items rather than single-item or pairwise labels; submitted a paper to **WSDM 2027**
  - Introduced a P-hot fixed-cardinality target representing multiple positive items per prediction; reduced the mismatch between single-item training objectives and full ranking-list evaluation
  - Adopted Gumbel-based Fenchel-Young loss (Gumbel-PFY) as the primary formulation for differentiable Top-K selection, which narrowed earlier algorithmic variants into one more focused claim
  - Evaluated the method against Bayesian Personalized Ranking (BPR), Binary Cross-Entropy (BCE), SoftmaxCE, and Talos objectives across Matrix Factorization (MF), LightGCN, and XSimGCL backbones to separate loss-level gains from encoder effects
  - Tested cardinality values from P=1 to P=5 at K=5, 10, and 20 cutoffs; found steady gains on Amazon Beauty and VideoGames but inconsistent trends on Gowalla, which demonstrated that the effect depends on dataset structure
  - Trained multiple GNN backbones on GPU-accelerated infrastructure; applied PyTorch Geometric's sparse matrix operations to handle large-scale user-item interaction graphs with millions of edges efficiently

Doubly Perturbed Optimizer for Gradient Estimation 07/2025 – Present

School of Mathematics and Physics, Department of Financial and Actuarial Mathematics | Advisor: Prof. Xiangqian Sun

- Developed a Doubly Perturbed Optimizer (DPO) framework in TensorFlow to enable exact sampling and stable gradient estimation for discrete latent variable models
  - Implemented dynamic programming with log-space prefix and suffix tables for exact subset selection; ensured accurate marginal probability calculations and numerical stability
  - Built a Discrete Variational Auto-Encoder (DVAE) on MNIST using multi-sample gradient estimation, KL warmup, and gradient clipping to prevent training collapse and cut estimation variance.
  - Conducted feature attribution on the Beer dataset using temperature annealing; achieved a test MSE of 0.0299 and 60.1% precision (at k=3) in identifying key explanatory features
  - Accelerated DVAE training by leveraging GPU parallelization for multi-sample gradient estimation, reducing per-epoch training time significantly compared to CPU-based sequential execution

Efficient Choice-Based Facility Location: A Hybrid AHP/DEA-Greedy Approach 04/2025 – 11/2025

School of Mathematics and Physics, Department of Financial and Actuarial Mathematics | Advisor: Prof. Xiangqian Sun

- Designed a hybrid AHP/DEA-Greedy optimization framework for competitive facility location modeling under customer choice dynamics
  - Built a utility-driven customer choice model with attraction-decay consideration sets to capture spatial competition and population distribution
  - Filtered out 40% of low-potential candidate sites prior to optimization by building an Analytic Hierarchy Process (AHP) and Data Envelopment Analysis (DEA) pre-screening pipeline
  - Accelerated large-scale solution search by 3.5x using a Greedy marginal-gain iterative algorithm; achieved 12% faster convergence than Simulated Annealing and Generalized Benders Decomposition

Multi-task Spatio-Temporal Face Detection and Recognition System 06/2024 – 09/2024

School of Advanced Technology, Department of Computing | Advisor: Prof. Yushan Pan

- Built a spatio-temporal face detection and recognition pipeline combining Haar Cascade screening with deep residual CNNs for real-time video surveillance.
  - Cascaded a Haar Cascade detector upstream of the CNN architecture to prune non-face regions early and minimize unnecessary network operations
  - Fine-tuned deep residual feature extraction layers to improve model robustness under illumination variations and occlusions, reaching a 98.2% recognition accuracy
  - Evaluated multi-scale motion trajectory tracking under dynamic surveillance workloads; resulted in sustained real-time performance at 30+ FPS

PROFESSIONAL EXPERIENCE

Oriza Holdings **Suzhou, China**

Investment Research Assistant Intern 07/2025 – 08/2025

- Conducted market research and technological feasibility analysis in AI and Industrial Intelligence; mapped early-stage targets across LLM applications, industrial IoT, and automated visual inspection
  - Screened pitch decks and enterprise backgrounds; extracted technical moats such as proprietary fine-tuning datasets, model latency metrics, and domain-specific dataset quality to construct structured deal briefs
  - Evaluated unit economics and Gross Profit Margin (GPM) by analyzing deployment costs for RAG architectures; focused on vector database indexing and retrieval accuracy, as well as multi-agent workflow automation setups
  - Summarized technical evaluations and financial projections into internal investment memos and competitive landscape matrices to support the deal team's target evaluation process

PROJECT EXPERIENCE

Continuous-Time Smartphone Battery Drain Modeling | *COMAP 2026 Mathematical Contest in Modeling* 02/2026

- Constructed a continuous-time model that combined the coupled ODEs of electrical and thermal effects with mechanical power demand; achieved high-precision predictions of battery state of charge and multi-scenario battery life analysis
  - Built a multi-scenario telemetry dataset to evaluate Time-to-Empty (TTE) predictions across varying initial charge states; analyzed empirical discharge distributions across five standard user profiles to explain performance heterogeneity under identical battery levels
  - Validated predictive accuracy and robustness through 5-fold cross-validation on an 80/20 train-test split; showed high precision and model stability, particularly across medium- and high-power consumption workloads
  - Quantified battery degradation risks using a Continuous-Time Markov Chain (CTMC) state-switching model and capacity health factor

Dynamic Portfolio Adjustment and Macro Hedging Strategies | *Bloomberg 2024 Global Trading Challenge* 11/2024

- Implemented a quantitative risk control and asset allocation framework for the Bloomberg Trading Challenge; achieved a top 15% ROI in the Asia-Pacific region
  - Designed a dynamic position-sizing and automated risk-mitigation strategy using defined stop-loss and take-profit rules; reduced overall portfolio volatility by 30% during high-fluctuation trading periods
  - Combined multi-factor technical indicators with timing signals; achieved a 75% accuracy rate on trade execution signals through systematic trend analysis
  - Performed iterative backtesting workflows on historical trading data to refine execution parameters; led the strategy optimization cycle that secured a 91st-place finish out of 300+ competing regional teams

EXTRACURRICULAR ACTIVITIES

Peer Reviewer | *ACM CIKM 2026* 2026

- Reviewed three IR/recommendation papers on off-policy ranking optimization, differentiable Top-K operators, and scaling-law analysis of ranking metrics
- Assessed theoretical soundness, experimental validity, and industrial deployment evidence across bias-variance trade-offs, complexity/gradient analysis, and power-law scaling behavior

Guest Lecturer | *Calculus Lecture and Academic Outreach Practice* 09/2023-Present

- Independently designed and delivered calculus lectures; used case-based teaching to attract over 400 online and offline participants to attend

Member | *Academic Resource Integration and Learning Support Society* 02/2024-06/2024

- Responsible for writing the core lecture notes for calculus and linear algebra; planned and promoted academic resources throughout the entire school; achieved a download volume of over 5,000 times and an adoption rate of 80%

SKILLS

Software and Programming: Python | R | SQL | Java | C++ | MATLAB